

to attract additional buyers. This cost component can be effectively managed during implementation of an investment decision.

Permanent Impact represents the information cost component. The information content, whether real or perceived, causes market participants to adjust their prices to a new perceived fair value. The rationale is that informed investors typically buy undervalued stock and sell overvalued stock. As participants observe buy orders their perception (at least to some extent) is that the stock is undervalued and they will adjust their offer prices upwards. As participants observe sell orders their perception (again, at least to some extent) is that the stock is overvalued and they will adjust bid prices downward. It is an unavoidable trading cost.

Mathematically, we define market impact as the difference between the actual price trajectory after the order is released to the market and the price trajectory that would have occurred if the order were never released. Regrettably, we cannot observe both price trajectories simultaneously and it is not possible to construct a controlled experiment to measure both trajectories at the same time. As a result, market impact is often referred to as the *Heisenberg uncertainty principle of finance*.

Example 1: Temporary Market Impact

A trader receives a buy order for 50,000 shares of RLK. Market quotes show 1000 shares at \$50; 2000 shares at \$50.25; 3000 shares at \$50.50; and 4000 shares at \$50.75. The trader can only execute 1000 shares at the best available price and another 9000 shares at the higher prices for an average price of \$50.50. But this only represents 10,000 shares of the original 50,000 share order. In order to attract additional seller liquidity into the market, the trader must offer the market an incremental premium above \$50.75. The liquidity and urgency need of this trade caused the trader to incur impact. Another option available to the traders is to wait for additional sellers to arrive at the current market prices. If this occurs the trader will be able to transact at a better price, but if prices move higher due to general market movement while the trader is waiting for sellers to arrive the price could become even higher and the cost more expensive. Waiting for additional counterparties to arrive is always associated with market risk.

Example 2: Permanent Market Impact

A trader receives a buy order for 250,000 shares of RLK currently trading at \$50. However, inadvertently this information is released to the market signaling that the stock is undervalued. Thus, investors who currently own stock will be unwilling to sell shares at the undervalued price of \$50 and